

Trust-Aware Resilient Distributed Observer Design for Connected Vehicle Platoons

Q.H. Nguyen, Shengya Meng, H. Rafaralahy, M. Haddad, and A. Zemouche

Abstract—Connected vehicle platoons depend on distributed state estimation, but vehicle-to-vehicle data can be corrupted by falsified states, replayed or delayed packets, packet drops, and compromised vehicles. This paper proposes a trust-aware resilient distributed observer for vehicles described by a discrete kinematic bicycle model. The observer converts local and distributed consistency checks into online observer weights. The resulting observer builds a time-varying trusted-neighbor set while preserving a local-observer anchor, so unreliable data are down-weighted without discarding useful cooperation from reliable vehicles. Checkable local nonlinear input-to-state stability conditions are given for bounded model uncertainty, input mismatch, nonlinear remainders, bounded accepted attack values, and trust-induced switching along bounded trajectories. A finite-window rollback mechanism is also formulated to mitigate contamination caused by delayed trust decisions. Simulations on a five-vehicle platoon under position, speed, and packet-drop attacks show bounded estimation degradation; during the tested closed-loop attack interval, the trust-modulated adaptive and cooperative adaptive cruise-control scheme preserves positive inter-vehicle spacing through longitudinal projection.

Index Terms—Connected and Autonomous Vehicles, system state estimation, methods for security and privacy, distributed observers, trust management.

I. INTRODUCTION

AUTONOMOUS road vehicles are no longer only mechanical platforms with added electronics. Advances in onboard sensing, embedded computation, and AI-supported perception are turning them into mobile cyber-physical systems that process large sensor streams and close control loops in real time. However, single-vehicle autonomy is only one part of the emerging mobility architecture. The next step is cooperation: through vehicle-to-vehicle (V2V) and broader vehicle-to-everything (V2X) communication, multiple vehicles can share state information and coordinate maneuvers, enabling platooning, smoother traffic flow, and safer mobility. In a connected and automated vehicle platoon, each vehicle must therefore estimate not only its own state but also the states of surrounding platoon members from a mixture of onboard sensing and communicated information. This distributed esti-

mation layer is a critical component of cooperative adaptive cruise control and other platoon-level decision loops.

The same connectivity that enables cooperation also enlarges the attack surface. From a systems viewpoint, threats may enter through three main layers: the sensing and perception layer, where spoofed or degraded measurements can mislead the ego vehicle; the actuation and control layer, where manipulated control loops can create unsafe physical motion; and the communication layer, where denial-of-service, replay, delay, or false-data-injection attacks can corrupt the information exchanged between vehicles. The present paper focuses on this communication-driven vulnerability, while also treating faulty neighbor sensing or local estimation as unreliable reported data. This is not only a theoretical concern: cyberattacks and misbehavior in connected automated platoons and Internet-of-Vehicles networks have become practical research issues in intelligent transportation systems [?], [?], [?].

Once compromised information is accepted by a distributed observer, the resulting estimation error can propagate across the communication graph and degrade both safety and closed-loop performance. Conventional distributed observers often assume that exchanged data are reliable, which is restrictive for connected-vehicle applications exposed to cyberattacks, packet drops, and faulty nodes. A resilient observer should therefore decide online which neighbor information remains useful, reduce the influence of unreliable packets, and still preserve cooperation when the received data are consistent with the vehicle motion.

This paper develops a trust-aware distributed observer for resilient platoon state estimation. The main idea is to evaluate the behavioral consistency of received data, convert this consistency into trust scores, and use those scores to adapt the observer weights online. As a result, the observer can reduce the influence of unreliable neighbors while preserving useful cooperation from reliable vehicles. The proposed formulation clarifies the threat model, separates the trust-scoring layer from the observer conditions needed for ISS analysis, introduces a bounded rollback module for delayed decisions, and evaluates the estimator under multiple communication faults.

The main contributions are summarized as follows:

- A resilient distributed state-estimation architecture for connected vehicle platoons under adversarial communication.
- A behavioral trust framework that combines local consistency, distributed-estimate consistency, freshness, and temporal smoothing to construct a trusted-neighbor set.
- A trust-adaptive observer-weight design with admissible mixing coefficients and checkable local nonlinear input-

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QCar/LIMO implementation resources are available at https://github.com/kslhuy/QCar2_Cran; the demonstration video is available at <https://youtu.be/t6-soznp6ks>.

to-state stability (ISS) conditions for bounded modeling, input, attack, and nonlinear mismatch terms in the bicycle-error dynamics.

- A finite-window rollback mechanism, stated as an implementable replay procedure, that can re-propagate recent observer updates after delayed trust decisions.
- Simulation evidence under position, velocity, and packet-drop attack cases, together with a closed-loop spacing check during the attack interval using the current ACC/CACC structure through path-longitudinal projection.

The remainder of the paper is organized as follows. Section ?? reviews related work and motivates the proposed approach. Section ?? introduces the graph notation, vehicle model, and threat model. Section ?? presents the distributed observer architecture. Section ?? defines the trust framework. Section ?? develops the trust-based weight design and stability analysis. Section ?? presents the delayed-decision rollback module. Section ?? reports the simulation study, QCar/LIMO platform implementation, and controller spacing check, and Section ?? concludes the paper.

II. RELATED WORK AND MOTIVATION

Security and resilience for connected vehicle platoons have been studied through attack detection, fault-tolerant estimation, and mitigation-oriented control. Observer-based methods compare model-predicted states with measurements or received data to detect abnormal behavior [?], [?], [?], [?]. These methods are attractive because they are model-aware and can be implemented locally, but their performance degrades when the observer itself uses corrupted neighbor information.

Consensus and distributed-estimation methods exploit redundancy across multiple agents or sensors [?], [?], [?]. They can improve robustness by cross-validating information, but many approaches implicitly require sufficiently reliable neighbors or fixed communication weights. Under coordinated attacks, persistent packet loss, or biased local observers, fixed consensus weights can spread wrong estimates instead of isolating them. More general secure-estimation methods include bank-of-observers selection under sparse sensor attacks [?] and distributed ℓ_1 state-and-fault estimation for multi-agent

systems [?]; these provide useful resilient-estimation baselines, but they do not directly provide trust-adaptive V2V observer weights for platoons.

Set-membership and data-driven techniques provide another line of resilient estimation and detection. Zonotopic set-based estimators propagate all states consistent with bounded uncertainties and can guarantee true-state containment under time-varying sensor attacks when sufficient uncompromised sensing remains [?]. Data-driven attack-detection methods instead use attack-free input-output trajectories to identify uncompromised sensors online, including extensions for replay and delay attacks [?]. These methods are useful for guaranteed or model-learning-based security analysis, but they are generally centralized or detection-oriented and do not directly adapt the distributed V2V observer weights used by a platoon controller.

Trust-based approaches assign reliability scores to agents or data streams. Physical-layer and connectivity-based methods use signal fingerprints or communication characteristics to authenticate neighbors [?], [?], while behavior-based methods compare exchanged data with local sensing or expected motion [?], [?]. Context-aware IoV trust management further incorporates application context, rational trust-attribute weighting, and adaptive misbehavior thresholds to improve attack resistance in dynamic vehicular networks [?]. Trust can also be injected into control policies to reduce the influence of unreliable vehicles [?], [?]. However, these trust mechanisms are usually diagnostic, reputation-oriented, or controller-side filters; they do not directly provide a trust-adaptive distributed observer with stability guarantees and delayed-decision rollback.

The present work is motivated by this gap. Instead of treating trust as an external diagnostic signal, the proposed method feeds trust directly into the observer-weight design. The observer therefore adapts its communication graph online while preserving a local-observer anchor, which improves recoverability after transient attacks and supports a stability analysis for bounded disturbances.

Notations: For vectors $z_1, \dots, z_N$, $\text{col}(z_1, \dots, z_N)$ denotes their vertical concatenation, and $\rho(M)$ denotes the spectral radius of a square matrix M . $\mathbf{1}_N$ is the all-ones vector.

TABLE I
POSITIONING OF THE PROPOSED METHOD RELATIVE TO RELATED RESILIENT PLATOON/STATE-ESTIMATION APPROACHES.

Approach	Typical attack class	Dist. observer	Adaptive trust weights	ISS/stability	Rollback	Closed-loop use
Observer residual detection [?], [?], [?], [?]	Sensor/communication anomalies	Partial	No	Case dependent	No	Partial
Consensus/distributed estimation [?], [?], [?]	Noise and model mismatch	Yes	No	Yes	No	Limited
Bank-of-observers secure estimation [?]	Sparse sensor attacks	No (bank)	No	Yes	No	No
Distributed ℓ_1 state/fault estimation [?]	Sparse faulty agents	Yes	No	Error bound	No	No
Set-membership/zonotopic estimation [?]	Time-varying sensor attacks	No	No	Set inclusion	No	No
Data-driven attack detection [?]	Sensor, replay, delay attacks	No	No	Feasibility cond.	No	No
Trust/context-aware reputation filtering [?], [?], [?], [?], [?], [?], [?]	Misbehavior and unreliable links	Partial	Partial	Limited	No	Partial
Proposed trust-aware observer	Bias, intermittent faults, replay/delay, packet drops	Yes	Yes	Yes	Optional	Yes

III. PROBLEM FORMULATION

In this section, a group of connected automated vehicles traveling on a two-dimensional plane are considered. Each vehicle can measure its own state through onboard sensors and exchange state information with neighboring vehicles via vehicle-to-vehicle (V2V) communication. Graph theory is used to characterize the communication topology among vehicles, while a discrete-time kinematic bicycle model is adopted to describe the motion dynamics of each vehicle. Considering potential threats such as data falsification, replay attacks, communication-delay attacks, denial-of-service (DoS) attacks, and erroneous neighbor state reports caused by sensor faults, we aim to design a resilient distributed state observer that enables each vehicle to estimate the state of the entire platoon under unreliable communication while ensuring bounded estimation errors.

A. Kinematic Bicycle Vehicle Model with V2V Communication

To describe the dynamics of each vehicle, we use a discrete-time kinematic bicycle model. The state and input of vehicle i are defined as

$$x_i = [X_i \ Y_i \ \psi_i \ v_i]^\top, \quad u_i = [a_i \ \delta_i]^\top,$$

where (X_i, Y_i) is the planar position, ψ_i is the yaw angle, v_i is the longitudinal speed, a_i is the acceleration, and δ_i is the front steering angle. With sampling time T_s and wheelbase L_w , the nominal forward-Euler model is

$$\begin{aligned} x_i(t+1) &= f_b(x_i(t), u_i(t)) + \Delta_i(t), \\ y_i(t) &= h_i(x_i(t)), \end{aligned} \quad (1)$$

where $\Delta_i(t)$ collects modeling mismatch, discretization error, and external disturbances, and

$$f_b(x_i, u_i) = \begin{bmatrix} X_i + T_s v_i \cos \psi_i \\ Y_i + T_s v_i \sin \psi_i \\ \psi_i + \frac{T_s v_i}{L_w} \tan \delta_i \\ v_i + T_s a_i \end{bmatrix}. \quad (2)$$

For the full-state ego-measurement case used in the observer design, $h_i(x_i) = x_i$. If a reduced sensor suite is used, h_i denotes the corresponding local measurement map. Throughout the observer and analysis, state differences are denoted by ordinary subtraction. For the yaw component, the difference is interpreted in local angle coordinates and wrapped with $\text{wrap}_\pi(\cdot)$ when needed.

Let $x = \text{col}(x_1, \dots, x_N)$, $u = \text{col}(u_1, \dots, u_N)$, $\Delta = \text{col}(\Delta_1, \dots, \Delta_N)$, and $f_N(x, u) = \text{col}(f_b(x_1, u_1), \dots, f_b(x_N, u_N))$. The overall vehicle dynamics are

$$\begin{cases} x(t+1) = f_N(x(t), u(t)) + \Delta(t), \\ y_i(t) = h_i(x_i(t)), \end{cases} \quad (3)$$

B. Communication graph

The V2V communication layer is described by a graph $\mathcal{G} = \{\mathcal{V}, \mathcal{E}, \mathcal{A}\}$, where $\mathcal{V} = \{1, 2, \dots, N\}$ is the set of vehicles, $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ is the edge set describing the communication

topology, and $\mathcal{A} = [a_{ij}] \in \mathbb{R}^{N \times N}$ is the adjacency matrix. The neighbor set of vehicle i is

$$\mathcal{N}_i = \{j \in \mathcal{V} \mid (j, i) \in \mathcal{E}\}. \quad (4)$$

The Laplacian matrix $\mathcal{L} = [l_{ij}]$ is defined by

$$l_{ij} = \begin{cases} \sum_{k=1}^N a_{ik}, & i = j, \\ -a_{ij}, & i \neq j. \end{cases} \quad (5)$$

Although the communication layer may be undirected, each vehicle can independently adjust its trust-based communication weights toward neighboring vehicles under potential attacks. Because such trust weights are generally asymmetric, the effective information flow becomes directional. Thus, the trust-weighted virtual communication graph can be modeled as a directed weighted graph. More details about the trust framework and its influence on the observer weights are provided in Section ??.

C. Problem Formulation and Threat Model

Each vehicle estimates the full platoon state using onboard measurements and packets received from neighboring vehicles. The objective is to ensure that the estimation error remains bounded and informative for platoon control even when a subset of received packets is unreliable. The proposed method does not require the host to know which channel is attacked or when the attack begins. It infers packet reliability online using physical consistency, agreement among distributed estimates, packet freshness, and trust smoothing.

We consider four communication and data-integrity faults:

- First, a false data injection (FDI) attack, also referred to as a data falsification attack, intentionally corrupts communicated state information by injecting an additive bias or substituting arbitrary values in selected fields such as planar position, speed, heading, acceleration, and steering. This manipulation typically preserves packet syntax while violating physical consistency and intervehicle agreement.
- Second, a replay attack resends previously transmitted valid packets, and a delay attack delivers otherwise valid packets with excessive latency. Both attacks yield stale but syntactically valid information that can pass basic format checks while contradicting the current motion evolution and heading consistency.
- Third, a denial of service (DoS) attack prevents timely packet delivery by dropping packets, jamming the channel, or saturating the communication medium. The resulting effect is intermittent or persistent packet loss that can manifest as extended communication blackouts.
- Fourth, a benign neighbor fault, caused for example by sensor malfunction, calibration drift, or local estimator divergence, produces corrupted neighbor reported estimates that are physically inconsistent even without adversarial intent. Such faults can mimic integrity attacks at the packet level while arising from nonmalicious sources.

In the analysis, these effects are modeled as bounded disturbance terms, input mismatch terms, and linearization remainder terms, provided that the corresponding magnitudes are

bounded. Packet losses are addressed through packet freshness and the selection of trusted neighbors.

To make this bounded-input interpretation precise, every received packet is first passed through authentication, timestamp, and physical-plausibility gates before it can enter the observer. Denote by $z_l(t)$ the packet content transmitted by vehicle l and by $a_{i,l}(t)$ an additive corruption seen by host i . The value made available to the trust and observer layers is

$$\tilde{z}_{i,l}(t) = \Pi_{\Omega_z} (z_l(t - \tau_{i,l}(t)) + a_{i,l}(t)), \quad (6)$$

when the packet is authenticated and fresh; otherwise it is rejected and $\delta_{i,l}(t) = 0$. Here Π_{Ω_z} denotes projection or clipping onto a compact physically admissible packet set, including feasible position increments, speeds, headings, accelerations, steering angles, and timestamps. Therefore, an arbitrarily large injected value is either rejected by the validity gate or converted into a bounded accepted value. The ISS result below applies to these bounded accepted corruptions; it does not claim robustness to unbounded values injected directly into the observer without a receiving-side gate.

The attacker is assumed to affect V2V data received from compromised or faulty neighbors, but not the host vehicle's onboard measurement channel or the local bicycle model used by the observer. The method is designed to isolate unreliable incoming information and maintain platoon-level safety; it is not intended to reconstruct the exact true state of a malicious vehicle after that vehicle's data have been rejected unless independent reliable measurements remain available. This limitation is standard for distributed observer architectures that prioritize safe estimation over forensic reconstruction of compromised agents.

IV. DISTRIBUTED STATE OBSERVER ARCHITECTURE

This section introduces the observer architecture; the trust mechanism is detailed in Section ???. Two layers are used: a Local Observer ($\mathcal{LO}$) based on onboard sensors, and a Distributed Observer ($\mathcal{DO}$) that fuses information exchanged through communication links.

The notation distinguishes the physical origin of an estimate from the host that stores it. Vehicle j runs its own local observer $\mathcal{LO}_j$ from onboard measurements and broadcasts the corresponding anchor packet $\hat{x}_0^{(j)}$, together with timestamp and optional input information. It also broadcasts its fleet estimate $\hat{x}_j = \text{col}(\hat{x}_j^{(1)}, \dots, \hat{x}_j^{(N)})$. When host vehicle i uses $\hat{x}_0^{(j)}$ in $\mathcal{DO}_i^{(j)}$, this quantity is the local-observer output generated by target vehicle j : it is onboard information only when $i = j$, and it is a received V2V anchor packet when $i \neq j$. Therefore the anchor is not assumed to be automatically reliable; its availability and use are controlled by the validity indicator $\delta_{i,j}(t)$ and the anchor acceptance variable $\chi_{i0}^{(j)}(t)$ defined below.

A. General structure of the observer

The observer is defined by

$$(\mathcal{LO}_i) : \hat{x}_0^{(i)}(t+1) = f_b(\hat{x}_0^{(i)}(t), u_i(t)) + F_i(t)(y_i(t) - h_i(\hat{x}_0^{(i)}(t))) \quad (7a)$$

$$\begin{aligned} \hat{x}_{i,c}^{(j)}(t) &= \hat{x}_i^{(j)}(t) + \sum_{l \in \mathcal{N}_i} w_{il}^{(j)}(t)(\hat{x}_l^{(j)}(t) - \hat{x}_i^{(j)}(t)) \\ &\quad + w_{i0}^{(j)}(t)(\hat{x}_0^{(j)}(t) - \hat{x}_i^{(j)}(t)) \end{aligned} \quad (7b)$$

$$(\mathcal{DO}_i^{(j)}) : \hat{x}_i^{(j)}(t+1) = f_b(\hat{x}_{i,c}^{(j)}(t), \hat{u}_j(t)) \quad (7c)$$

where $\hat{x}_0^{(i)}(t)$ is the local-observer state of vehicle i , $F_i(t)$ is the corresponding time-varying gain, and $\mathcal{N}_i$ is the neighbor set of vehicle i in $\mathcal{G}$. The variable $\hat{x}_i^{(j)}(t)$ is vehicle i 's distributed estimate of vehicle j 's bicycle state. The intermediate state $\hat{x}_{i,c}^{(j)}(t)$ applies trust-weighted consensus and anchor corrections before the nonlinear bicycle prediction. The weights $w_{il}^{(j)}(t)$, $l \in \mathcal{N}_i$, and $w_{i0}^{(j)}(t)$ scale the neighbor-consensus and local-observer anchoring terms.

Remark 1. If a host forms $\hat{u}_j(t)$, for example from a known controller law or an input estimator for $\hat{a}_j(t)$ and $\hat{\delta}_j(t)$, the mismatch

$$\Delta u_j(t) = u_j(t) - \hat{u}_j(t) \quad (8)$$

will be treated as part of the disturbance vector.

The distributed observer $\mathcal{DO}_i^{(j)}$ estimates the state of vehicle $j \in \mathcal{V}$ from vehicle i . The neighbor-consensus term in (??) aligns $\hat{x}_i^{(j)}(t)$ with estimates received from $l \in \mathcal{N}_i$, while $w_{il}^{(j)}(t)$ sets each neighbor's influence. Since consensus alone may drive all estimates to a common biased value under disturbances or model mismatch, the anchoring term $w_{i0}^{(j)}(t)(\hat{x}_0^{(j)}(t) - \hat{x}_i^{(j)}(t))$ injects vehicle j 's target-local anchor as a reference before the bicycle prediction in (??).

B. Virtual communication graph

Each distributed observer $\mathcal{DO}_i^{(j)}$ needs reference information from the local observer $\mathcal{LO}_j$. To include this reference, we introduce a virtual anchor node "0" associated with the local observer $\mathcal{LO}_j$. At time t , the active weighted virtual graph $\bar{\mathcal{G}}^{(j)}(t) = \{\bar{\mathcal{V}}, \bar{\mathcal{E}}^{(j)}(t)\}$ is

$$\begin{aligned} \bar{\mathcal{V}} &= \mathcal{V} \cup \{0\}, \\ \bar{\mathcal{E}}^{(j)}(t) &= \{(l, i) \mid l \in \mathcal{N}_i, w_{il}^{(j)}(t) > 0\} \\ &\quad \cup \{(0, i) \mid w_{i0}^{(j)}(t) > 0\}. \end{aligned} \quad (9)$$

Let $W^{(j)}(t) = [w_{il}^{(j)}(t)]_{i,l=1}^N$, with $w_{il}^{(j)}(t) = 0$ when $l \notin \mathcal{N}_i$, and let $w_0^{(j)}(t) = \text{col}(w_{10}^{(j)}(t), \dots, w_{N0}^{(j)}(t))$. The corresponding weighted adjacency matrix can be written as

$$\bar{\mathcal{A}}^{(j)}(t) = \begin{bmatrix} 0 & 0 \\ w_0^{(j)}(t) & W^{(j)}(t) \end{bmatrix}. \quad (10)$$

In the error dynamics, the anchor node is not a state to be estimated. We therefore use the grounded Laplacian $\bar{\mathcal{L}}^{(j)}(t) = [\bar{l}_{ik}^{(j)}(t)] \in \mathbb{R}^{N \times N}$ obtained by removing the anchor row and column while retaining the anchor weights in the diagonal:

$$\begin{aligned} \bar{l}_{ii}^{(j)}(t) &= \sum_{l \in \mathcal{N}_i \cup \{0\}} w_{il}^{(j)}(t), \\ \bar{l}_{ik}^{(j)}(t) &= \begin{cases} -w_{ik}^{(j)}(t), & k \neq i, k \in \mathcal{N}_i, \\ 0, & k \neq i, k \notin \mathcal{N}_i. \end{cases} \end{aligned} \quad (11)$$

Thus, $I_N - \bar{\mathcal{L}}^{(j)}(t)$ contains the residual self weight on the diagonal and the neighbor weights off diagonal, while the missing row mass $w_{i0}^{(j)}(t)$ enters separately through the local-anchor input $w_0^{(j)}(t) \otimes A_j(t)$ in the error recursion. This virtual graph combines local estimation through node 0 and neighbor consensus through $\mathcal{G}$, so each local observer can act as an anchor for distributed estimation.

C. Main objectives

Fixed observer weights assume reliable data exchange, which is not valid when compromised vehicles transmit falsified information. The objective is to define a trust measure for received data and use it to update $w_{il}^{(j)}(t)$ and $w_{i0}^{(j)}(t)$ online, while choosing the target-local gain $F_j(t)$ so the local nonlinear bicycle-error synthesis in Section ?? yields an ISS convergence bound. In particular, there exist constants $c_0, c_1 > 0$ and $\lambda \in (0, 1)$ such that

$$\|e_0^{(j)}(t)\| \leq c_0 \lambda^t \|e_0^{(j)}(0)\| + c_0 \bar{d}_0, \quad (12)$$

$$\|e_{\mathcal{DO}}^{(j)}(t)\| \leq c_1 \lambda^t \|e_{\mathcal{DO}}^{(j)}(0)\| + c_1 \left(\|e_0^{(j)}(0)\| + \bar{d}_0 + \bar{\tilde{d}}_{\mathcal{DO}} \right). \quad (13)$$

where $e_0^{(j)}$ and $e_{\mathcal{DO}}^{(j)}$ denote the local and distributed estimation errors defined in Section ??, and $\bar{d}_0$, $\bar{d}_{\mathcal{DO}}$, and $\bar{\tilde{d}}_{\mathcal{DO}}$ are the bounded-input constants in Assumption ??.

V. TRUST SCORE FRAMEWORK

To support resilient estimation, each vehicle evaluates the reliability of received data. The proposed trust framework quantifies how much a host vehicle i trusts another vehicle l . Its structure is shown in Fig. ?. The formulas in this section define one implementable trust map. The stability result in Section ?? does not rely on a particular statistical optimality property of this map; it requires only that the resulting weights are nonnegative, bounded, and satisfy the stated contraction/connectivity checks. This separation is useful in practice: the trust layer can be replaced by another detector, while the observer analysis remains valid as long as the induced weights satisfy the same admissibility conditions.

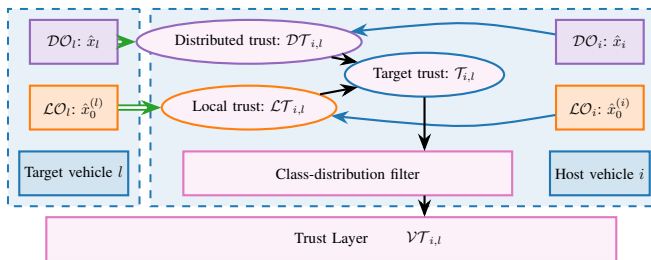


Fig. 1. Trust framework.

In this context, we define the following types of trust:

- **Local Trust** $\mathcal{LT}_{i,l}$: reliability of vehicle l 's local-observer data.
- **Distributed Trust** $\mathcal{DT}_{i,l}$: agreement between distributed estimates from l and those of i .

- **Target Trust** $\mathcal{T}_{i,l}$: instantaneous fusion of Local Trust and Distributed Trust.
- **Vehicle Trust** $\mathcal{VT}_{i,l}$: class-filtered trust score used by the observer.

The following subsections define these scores and their use in the observer.

A. Data validity indicator

Before trust computation, each received V2V packet is checked for authenticity and freshness:

$$\delta_{i,l}(t) = \begin{cases} 1, & \text{if the message is authenticated and fresh,} \\ 0, & \text{otherwise.} \end{cases} \quad (14)$$

Only packets with $\delta_{i,l}(t) = 1$ are used for trust evaluation.

B. Local trust

Local Trust $\mathcal{LT}_{i,l}$ measures the confidence of host vehicle i in the local-observer output of neighbor $l \in \mathcal{N}_i$. It uses bicycle-state consistency indicators: planar position, speed, heading, and an optional input-consistency check for acceleration and steering. Missing data are handled by a hold-and-decay rule.

1) *Planar position consistency*: Let $p_i = [X_i, Y_i]^\top$. The host transforms its onboard relative measurement of vehicle l into the global frame and denotes it by $r_{i,l}^{\text{meas}}(t) \in \mathbb{R}^2$. The relative position implied by the local observers is

$$r_{i,l}^{\text{obs}}(t) = \begin{bmatrix} \hat{X}_0^{(l)}(t) - \hat{X}_0^{(i)}(t) \\ \hat{Y}_0^{(l)}(t) - \hat{Y}_0^{(i)}(t) \end{bmatrix}.$$

The position-consistency score is

$$\iota_{i,l}^{\text{pos}} = \max \left(1 - \frac{\|r_{i,l}^{\text{obs}} - r_{i,l}^{\text{meas}}\|}{\varepsilon_p + \|r_{i,l}^{\text{meas}}\|}, 0 \right),$$

where $\varepsilon_p > 0$ avoids division by zero.

2) *Speed consistency*: The host compares the relative speed implied by the two local observers with the onboard measured relative speed $v_{i,l}^{\text{rel,meas}}(t)$:

$$\iota_{i,l}^{\text{speed}} = \max \left(1 - \frac{|\hat{v}_0^{(l)}(t) - \hat{v}_0^{(i)}(t) - v_{i,l}^{\text{rel,meas}}(t)|}{\varepsilon_v + |v_{i,l}^{\text{rel,meas}}(t)|}, 0 \right),$$

where $\varepsilon_v > 0$ avoids division by zero.

3) *Heading consistency*: Heading is checked using the reported yaw and a motion-derived yaw estimate. From two consecutive accepted position reports from vehicle l ,

$$\psi_l^{\text{pos}}(t) = \text{atan2}(\hat{Y}_0^{(l)}(t) - \hat{Y}_0^{(l)}(t-1), \hat{X}_0^{(l)}(t) - \hat{X}_0^{(l)}(t-1)).$$

If radar azimuth or LiDAR shape orientation is available, it can replace or complement $\psi_l^{\text{pos}}(t)$. The heading score is

$$\iota_{i,l}^{\text{heading}} = \max \left(1 - \frac{|\text{wrap}_\pi(\hat{\psi}_0^{(l)}(t) - \psi_l^{\text{pos}}(t))|}{\varepsilon_\psi + \psi_{\text{th}}}, 0 \right),$$

where $\psi_{\text{th}} > 0$ is a nominal heading tolerance. This check implements the heading-verification mechanism based on consecutive position reports or bearing measurements.

Assumption 1. When a vehicle is connected, it has access to basic information about its neighbors, including their type, length, and nominal wheelbase class.

4) *Optional input consistency:* If acceleration and steering commands are included in the received packet, the host checks them against finite-difference speed and yaw-rate estimates:

$$a_l^{\text{fd}}(t) = \frac{\hat{v}_0^{(l)}(t) - \hat{v}_0^{(l)}(t-n)}{nT_s},$$

$$\omega_l^{\text{bike}}(t) = \frac{\hat{v}_0^{(l)}(t)}{L_w} \tan \hat{\delta}_l(t).$$

With $\omega_l^{\text{fd}}(t) = \text{wrap}_\pi(\hat{\psi}_0^{(l)}(t) - \hat{\psi}_0^{(l)}(t-n))/(nT_s)$,

$$l_{i,l}^{\text{input}} = \max \left(1 - \frac{|\hat{a}_l - a_l^{\text{fd}}| + |\omega_l^{\text{bike}} - \omega_l^{\text{fd}}|}{\varepsilon_u + a_{\text{th}} + \omega_{\text{th}}}, 0 \right),$$

where a_{th} and ω_{th} are nominal acceleration and yaw-rate tolerances. If the command channels are not transmitted, $l_{i,l}^{\text{input}}$ is omitted from the fusion.

5) *Fusion and update rule:* For simplicity and reproducibility, the available indicators are fused by a weighted geometric product. Let $\omega_p, \omega_v, \omega_\psi, \omega_u \geq 0$ denote channel exponents. When the input channel is available, the local trust score is

$$\mathcal{LT}_{i,l}^{\text{raw}}(t) = \left(l_{i,l}^{\text{pos}}(t) \right)^{\omega_p} \left(l_{i,l}^{\text{speed}}(t) \right)^{\omega_v} \times \left(l_{i,l}^{\text{heading}}(t) \right)^{\omega_\psi} \left(l_{i,l}^{\text{input}}(t) \right)^{\omega_u}. \quad (15)$$

If the optional input channel is unavailable, the last factor is omitted. Missing or invalid packets are handled by

$$\mathcal{LT}_{i,l}(t) = \begin{cases} \mathcal{LT}_{i,l}^{\text{raw}}(t), & \text{if } \delta_{i,l}(t) = 1, \\ (1 - \lambda_h) \mathcal{LT}_{i,l}(t-1), & \text{otherwise} \end{cases} \quad (16)$$

where λ_h sets the decay rate during data loss.

Remark 2. Multiplicative fusion is conservative: one inconsistent channel is enough to reduce the local trust score. Larger exponents assign stronger influence to the corresponding consistency channel. The decay λ_h can increase with consecutive missed packets.

C. Distributed trust

Distributed Trust applies the same idea to exchanged distributed estimates. It is defined as

$$\mathcal{DT}_{i,l}(t) = \begin{cases} \gamma_{i,l}^{\text{self}}(t) \cdot \gamma_{i,l}^{\text{local}}(t) \cdot \gamma_{i,l}^{\text{host}}(t) & \text{if } \delta_{i,l}(t) = 1 \\ (1 - \lambda_h) \mathcal{DT}_{i,l}(t-1), & \text{otherwise} \end{cases} \quad (17)$$

where $\gamma_{i,l}^{\text{self}}$, $\gamma_{i,l}^{\text{local}}$, and $\gamma_{i,l}^{\text{host}}$ check consistency with the sender's local observer, the host's distributed estimate, and the host's local measurements, respectively.

1) *Sender self-consistency:* The sender's distributed estimate of its own state should remain close to its local observer state:

$$\gamma_{i,l}^{\text{self}}(t) = \exp \left(- \left(\hat{x}_l^{(l)}(t) - \hat{x}_0^{(l)}(t) \right)^\top \Sigma_{\text{self}}^{-1} \left(\hat{x}_l^{(l)}(t) - \hat{x}_0^{(l)}(t) \right) \right), \quad (18)$$

where Σ_{self} is the covariance matrix associated with the sender self-consistency residual.

2) *Host-estimate consistency:* The host compares the estimate received from target l with its own estimate using the Mahalanobis distance:

$$\rho_{i,l}^{(j)}(t) = \left(\hat{x}_i^{(j)}(t) - \hat{x}_l^{(j)}(t) \right)^\top \Sigma_{\text{host}}^{-1} \left(\hat{x}_i^{(j)}(t) - \hat{x}_l^{(j)}(t) \right) \quad (19)$$

where Σ_{host} is the diagonal covariance matrix of state-channel discrepancies:

$$\Sigma_{\text{host}} = \text{diag} \left(\sigma_1^2, \sigma_2^2, \dots, \sigma_n^2 \right), \quad (20)$$

where n is the number of states and σ_k^2 is the variance of the k th state discrepancy.

The total discrepancy score is

$$D_{i,l}^{\text{host}}(t) = \sum_{j=1}^N \rho_{i,l}^{(j)}(t). \quad (21)$$

A small $D_{i,l}^{\text{host}}(t)$ means that vehicle l 's estimates are close to the host's estimates. The corresponding trust factor is

$$\gamma_{i,l}^{\text{host}}(t) = \exp \left(-D_{i,l}^{\text{host}}(t) \right). \quad (22)$$

where $\gamma_{i,l}^{\text{host}}(t) \in (0, 1]$.

3) *Consistency with host's local measurement:* Host vehicle i uses relative measurements to nearby vehicles, e.g., planar relative position, range, bearing, relative speed, or heading difference, to check whether the estimates from target l are consistent with local sensing.

Let $y_{i,j}(t) \in \mathbb{R}^{m_{ij}}$ be the relative measurement from vehicle i to vehicle j , e.g., the global-frame relative position $p_j(t) - p_i(t)$ or a range-bearing vector transformed through the host pose.

The estimate received from target l should approximately satisfy

$$H \left(\hat{x}_l^{(j)}(t) - \hat{x}_l^{(i)}(t) \right) \approx y_{i,j}(t).$$

where $H \in \mathbb{R}^{m_{ij} \times n}$ extracts the measured states.

The consistency error is

$$\varepsilon_{i,l}^{(j)}(t) = H \left(\hat{x}_l^{(j)}(t) - \hat{x}_l^{(i)}(t) \right) - y_{i,j}(t), \quad (23)$$

Using Σ_{local} as in (??), define

$$E_{i,l}^{(j)}(t) = \left(\varepsilon_{i,l}^{(j)}(t) \right)^\top \Sigma_{\text{local}}^{-1} \varepsilon_{i,l}^{(j)}(t). \quad (24)$$

The total consistency error is

$$E_{i,l}(t) = \sum_{j \in \mathcal{N}_i} E_{i,l}^{(j)}(t). \quad (25)$$

A higher $E_{i,l}(t)$ means that the estimates from target l disagree with local relative measurements.

The trust factor is

$$\gamma_{i,l}^{\text{local}}(t) = \exp \left(-E_{i,l}(t) \right), \quad (26)$$

where $\gamma_{i,l}^{\text{local}}(t) \in (0, 1]$.

Remark 3. The covariance matrices Σ_{host} and Σ_{local} can be estimated from a recent nominal-data window at vehicle i , avoiding covariance transmission between nodes.

D. Target trust and temporal evolution

We combine the Local Trust and Distributed Trust into a single instantaneous trust score for target l :

$$\mathcal{T}_{i,l}(t) = \mathcal{LT}_{i,l}(t) \cdot \mathcal{DT}_{i,l}(t). \quad (27)$$

$\mathcal{T}_{i,l}(t)$ measures the instantaneous quality of target l 's data. Because this value can vary abruptly, a class-distribution temporal filter is used instead of applying the instantaneous scalar directly. Let $q = 5$ denote the number of ordered trust classes: *Unreliable*, *Poor*, *Acceptable*, *Good*, and *Excellent*. Define the class scores

$$\rho_m = \frac{m-1}{q-1}, \quad m = 1, \dots, q, \quad (28)$$

and the corresponding intervals

$$\mathcal{I}_m = \begin{cases} \left[\frac{m-1}{q}, \frac{m}{q} \right), & m = 1, \dots, q-1, \\ \left[\frac{q-1}{q}, 1 \right], & m = q. \end{cases} \quad (29)$$

The instantaneous trust value is discretized into a one-hot class vector

$$r_{i,l,m}(t) = \begin{cases} 1, & \mathcal{T}_{i,l}(t) \in \mathcal{I}_m, \\ 0, & \text{otherwise,} \end{cases} \quad r_{i,l}(t) \in \mathbb{R}^q. \quad (30)$$

For example, $\mathcal{T}_{i,l}(t) = 0.72$ belongs to the *Good* class for $q = 5$, so $r_{i,l}(t) = [0, 0, 0, 1, 0]^\top$.

The accumulated class evidence is updated as

$$R_{i,l}(t+1) = (1 - \beta_{\text{trust}} \mathcal{T}_{i,l}(t)) R_{i,l}(t) + r_{i,l}(t), \quad (31)$$

where $R_{i,l}(t) \in \mathbb{R}_{\geq 0}^q$ is initialized as $R_{i,l}(0) = 0$ unless prior interaction evidence is available, and $\beta_{\text{trust}} \in [0, 1]$ controls the decay of past class evidence. The modulation by $\mathcal{T}_{i,l}(t)$ prevents one isolated low-quality sample from immediately erasing the accumulated history, while repeated low-quality samples keep adding evidence to the lower classes. To regularize the class distribution, let $\alpha_{\text{tr}} \in \mathbb{R}_{\geq 0}^q$ be a prior trust-class distribution satisfying $\sum_{m=1}^q \alpha_{\text{tr},m} = 1$, and let $C_{\text{tr}} > 0$ be a confidence parameter. The prior α_{tr} represents the initial belief about vehicle l before enough packets are observed; if no prior information is available, one can use the uniform choice $\alpha_{\text{tr},m} = 1/q$. The parameter C_{tr} controls how strongly this prior affects the estimate: larger C_{tr} gives more weight to the prior, while smaller C_{tr} lets online evidence dominate faster. The normalized trust-class distribution is

$$S_{i,l,m}(t) = \frac{R_{i,l,m}(t) + C_{\text{tr}} \alpha_{\text{tr},m}}{C_{\text{tr}} + \sum_{s=1}^q R_{i,l,s}(t)}, \quad m = 1, \dots, q. \quad (32)$$

The smoothed vehicle trust used by the observer is the expected trust level

$$\mathcal{VT}_{i,l}(t) = \sum_{m=1}^q \rho_m S_{i,l,m}(t). \quad (33)$$

A larger β_{trust} gives a more reactive trust history, while a smaller value gives smoother trust evolution. The confidence C_{tr} controls how strongly the prior influences the estimate when only a short interaction history is available. The observer uses $\mathcal{VT}_{i,l}(t)$ to weigh information received from vehicle l .

E. Practical parameter selection

The trust parameters are selected from nominal data and physical limits rather than tuned directly on attack cases. First, the receiver defines the admissible packet set Ω_z in (??) from vehicle limits, road geometry, timestamp tolerance, maximum speed, acceleration, steering angle, and feasible inter-sample motion. Second, the covariance matrices Σ_{self} , Σ_{host} , and Σ_{local} are estimated from an attack-free calibration window and regularized with positive lower bounds on every diagonal entry. Third, the exponents $\omega_p, \omega_v, \omega_\psi, \omega_u$ encode channel confidence: a channel with lower nominal noise or higher safety relevance receives a larger exponent, while an unavailable channel is removed from the product. Fourth, the threshold $\theta_{\min}$ is chosen from nominal validation data so that benign packets are rarely rejected, then checked under representative faults. Finally, β_{trust} and C_{tr} set the detection–recovery trade-off: larger β_{trust} reacts faster to repeated low-trust evidence, while larger C_{tr} prevents short histories from dominating the prior. After these values are fixed, the observer weights are checked against the stability conditions in Section ??.

VI. TRUST-BASED OBSERVER WEIGHT DESIGN

The trust framework provides each vehicle with smoothed vehicle-trust scores $\mathcal{VT}_{i,l}(t)$ representing the reliability of information from other vehicles. This section shows how these values are used to define adaptive observer weights $w_{i,l}^{(j)}(t)$ and establishes conditions for stability of the estimation error dynamics.

A. Weight based on the trust

Using the virtual graph $\bar{\mathcal{G}}^{(j)}$, each vehicle i first selects the trusted neighbor sources at time t :

$$\mathcal{LN}_i^{(j)}(t) = \left\{ l \in \mathcal{N}_i \mid \mathcal{VT}_{i,l}(t) \geq \theta_{\min} \right\}. \quad (34)$$

The selected trust scores are then used directly as reliability scores,

$$s_{i,l}^{(j)}(t) = \begin{cases} \mathcal{VT}_{i,l}(t), & l \in \mathcal{LN}_i^{(j)}(t), \\ 0, & \text{otherwise,} \end{cases} \quad (35)$$

$$S_i^{(j)}(t) = \epsilon + \sum_{r \in \mathcal{N}_i} s_{i,r}^{(j)}(t), \quad (36)$$

where $\epsilon > 0$ avoids division by zero. Let $\eta_0, \eta_N \in [0, 1]$ denote the local-anchor and neighbor-consensus budgets, respectively, with

$$\eta_0 + \eta_N \leq 1. \quad (37)$$

Let $\chi_{i0}^{(j)}(t) \in \{0, 1\}$ indicate that the local-anchor packet $\hat{x}_0^{(j)}(t)$ is available and accepted at host i . For the target's own host, $\chi_{j0}^{(j)}(t) = 1$ and $\mathcal{LT}_{j,j}(t) = 1$; if the anchor packet is unavailable or rejected, $\chi_{i0}^{(j)}(t) = 0$ and the anchor weight is zero. The local-anchor weight is

$$w_{i0}^{(j)}(t) = \begin{cases} \eta_0 \chi_{i0}^{(j)}(t) \mathcal{LT}_{i,j}(t), & \chi_{i0}^{(j)}(t) = 1, \\ 0, & \mathcal{LT}_{i,j}(t) \geq \theta_{\min}, \\ & \text{otherwise,} \end{cases} \quad (38)$$

and the neighbor weights are

$$w_{il}^{(j)}(t) = \eta_N \frac{s_{i,l}^{(j)}(t)}{S_i^{(j)}(t)}, \quad l \in \mathcal{N}_i. \quad (39)$$

Thus, rejected neighbors receive zero weight, while accepted neighbors share the fixed budget η_N in proportion to their trust values. The remaining mass is the implicit self-prediction weight

$$w_{is}^{(j)}(t) = 1 - w_{i0}^{(j)}(t) - \sum_{l \in \mathcal{N}_i} w_{il}^{(j)}(t). \quad (40)$$

The following lemma shows that the proposed trust-based weight law always generates admissible mixing coefficients. In particular, the correction step is a nonnegative weighted combination of the self prediction, trusted neighbors, and the local-observer anchor. This property is important because it prevents the observer from extrapolating unreliable information and guarantees that bounded attacks enter the error dynamics only through bounded weighted inputs.

Lemma 1 (Admissibility of trust weights). *The weight law (??)–(??) satisfies $w_{il}^{(j)}(t) \geq 0$, $w_{i0}^{(j)}(t) \geq 0$, and*

$$\sum_{l \in \mathcal{N}_i \cup \{0\}} w_{il}^{(j)}(t) \leq 1 \quad (41)$$

for all i, j and all t .

Proof. By construction, $s_{i,l}^{(j)}(t) \geq 0$ and $S_i^{(j)}(t) > 0$, hence $w_{il}^{(j)}(t) \geq 0$ for all $l \in \mathcal{N}_i$. Also, $0 \leq w_{i0}^{(j)}(t) \leq \eta_0$ because $\chi_{i0}^{(j)}(t) \in \{0, 1\}$ and $\mathcal{LT}_{i,j}(t) \in [0, 1]$. The neighbor contribution satisfies

$$\sum_{l \in \mathcal{N}_i} w_{il}^{(j)}(t) = \eta_N \frac{\sum_{l \in \mathcal{N}_i} s_{i,l}^{(j)}(t)}{\epsilon + \sum_{l \in \mathcal{N}_i} s_{i,l}^{(j)}(t)} \leq \eta_N.$$

Therefore,

$$\sum_{l \in \mathcal{N}_i \cup \{0\}} w_{il}^{(j)}(t) = w_{i0}^{(j)}(t) + \sum_{l \in \mathcal{N}_i} w_{il}^{(j)}(t) \leq \eta_0 + \eta_N \leq 1. \quad \blacksquare$$

The lemma provides bounded mixing, but bounded mixing alone is not sufficient for convergence because the bicycle model contains prediction dynamics. The accepted trust-induced weights must also provide enough local-anchor mass, or enough rooted multi-step information flow, to contract the distributed prediction error. Let

$$M^{(j)}(t) = I_N - \bar{\mathcal{L}}^{(j)}(t). \quad (42)$$

By Lemma ??, $M^{(j)}(t)$ is nonnegative and its i th row sum is $1 - w_{i0}^{(j)}(t)$. Therefore, in the infinity norm,

$$\|M^{(j)}(t)\|_\infty = \max_{i \in \mathcal{V}} (1 - w_{i0}^{(j)}(t)). \quad (43)$$

The proposed weight law gives a direct lower bound on this missing row mass whenever the target-local anchor packet is accepted.

Lemma 2 (Anchor-induced one-step contraction). *Suppose that, for target j , the accepted anchor packet is available to every host at time t and satisfies $\chi_{i0}^{(j)}(t) = 1$ and*

$\mathcal{LT}_{i,j}(t) \geq \theta_{\min}$ for all $i \in \mathcal{V}$. Then the weight law (??) gives

$$w_{i0}^{(j)}(t) \geq \underline{w}_0, \quad \underline{w}_0 := \eta_0 \theta_{\min}. \quad (44)$$

Consequently,

$$\|M^{(j)}(t)\|_\infty \leq 1 - \underline{w}_0. \quad (45)$$

If the bicycle map is Lipschitz on the admissible compact region in the same state norm with constant $L_{x,j}$ and

$$L_{x,j}(1 - \underline{w}_0) < 1, \quad (46)$$

then the homogeneous distributed prediction error is a one-step contraction.

Proof. Under the stated acceptance condition, (??) gives $w_{i0}^{(j)}(t) = \eta_0 \mathcal{LT}_{i,j}(t) \geq \eta_0 \theta_{\min}$. The row-sum identity (??) then gives $\|M^{(j)}(t)\|_\infty \leq 1 - \underline{w}_0$. Combining this with (??) yields the contraction factor in (??). $\blacksquare$

When direct anchors are intermittent, the same argument must be applied over a window. These weights induce a switching communication graph because the trusted-neighbor set changes online. A plain union spanning tree is not enough unless the anchor information can move through the matrix product in time order. We therefore use the following sequential rooted accepted-graph condition, where an edge $l \rightarrow i$ means that the accepted weight $w_{il}^{(j)}$ is positive.

Assumption 2 (Sequentially rooted accepted graph). *There exist an integer $T_c > 0$ and a scalar $\underline{w} > 0$ such that, for every target j , every start time $t \geq 0$, and every host $i \in \mathcal{V}$, there are nodes $v_0 = 0, v_1, \dots, v_m = i$ and times $t \leq \tau_1 < \dots < \tau_m \leq t + T_c - 1$ for which the accepted edge $v_{r-1} \rightarrow v_r$ is present in $\bar{\mathcal{G}}^{(j)}(\tau_r)$. Each accepted edge weight on these temporal paths is at least $\underline{w}$, and each implicit self-prediction weight used to carry information between two consecutive path times is also at least $\underline{w}$. Moreover, if the path reaches the destination host i at a time $\tau_m < t + T_c - 1$, then the implicit self-prediction weight of host i is also at least $\underline{w}$ at every remaining instant $\tau_m < \tau \leq t + T_c - 1$, so that the anchor-induced row-sum deficit at host i is preserved until the end of the window.*

Assumption ?? means that every host estimate of target j receives anchor information either directly or through trusted intermediate estimates within each window. This is stronger than ordinary connectivity: the root must be the target-local observer node, the links must be accepted by the trust layer, and the path must be ordered along the product. A standard jointly rooted union condition implies this assumption after enlarging the window when accepted links recur in every joint-connectivity interval and the implicit self-prediction weights are uniformly positive; otherwise the product condition below must be verified directly.

Lemma 3 (Rooted block contraction). *If Assumption ?? holds, then the grounded trust-mixing product satisfies*

$$\|M^{(j)}(t + T_c - 1) \cdots M^{(j)}(t)\|_\infty \leq 1 - \underline{w}^{T_c}. \quad (47)$$

Let $G_{x,j}(T_c)$ be an upper bound, in the chosen state norm, on the nonlinear bicycle prediction gain over any admissible T_c -step block. Because the one-step bicycle prediction is Lipschitz with constant $L_{x,j}$ in this norm on the compact admissible region, the T_c -step prediction gain satisfies

$$G_{x,j}(T_c) \leq L_{x,j}^{T_c}. \quad (48)$$

Since $L_{x,j}$ is scalar, it factors out of the matrix product. In particular, for the componentwise error-size vector

$$E^{(j)}(t) = \text{col} \left(\|e_1^{(j)}(t)\|_*, \dots, \|e_N^{(j)}(t)\|_* \right), \quad (49)$$

the homogeneous recursion satisfies, componentwise,

$$E^{(j)}(t + T_c) \leq L_{x,j}^{T_c} (M^{(j)}(t + T_c - 1) \cdots M^{(j)}(t)) E^{(j)}(t). \quad (50)$$

This explains why the prediction gain and the trust-mixing product separate even though prediction and mixing are interleaved in time. If

$$G_{x,j}(T_c)(1 - \underline{w}^{T_c}) \leq \alpha_c < 1, \quad (51)$$

then the distributed homogeneous error is contractive over every window of length T_c . The one-step condition (??) is the special case $T_c = 1$ with direct anchor acceptance at every step.

Proof. The matrices $M^{(j)}(\tau)$ are nonnegative and substochastic. A positive anchor weight creates a strict row-sum deficit in the row receiving that anchor. For any host i , Assumption ?? gives a temporal accepted path from the anchor to i . Each accepted edge transmits at least a factor $\underline{w}$ of the existing deficit to the next node, and each waiting step preserves at least a factor $\underline{w}$ through the implicit self-prediction weight. Since the path transmissions and waiting steps occupy no more than T_c time instants, the terminal-waiting clause of Assumption ?? guarantees that, once the deficit reaches host i at time τ_m , it is carried by a factor of at least $\underline{w}$ per step up to $t + T_c - 1$. Hence it appears in the i th row sum of the full product, which is at most $1 - \underline{w}^{T_c}$. Taking the maximum row sum gives (??). Multiplying by the bounded T_c -step bicycle prediction gain gives (??). ■

Thus, contraction is not assumed from trust scoring alone. The weight law supplies the anchor lower bound through $\eta_0 \theta_{\min}$ when anchor packets are directly accepted, and Assumption ?? supplies the corresponding lower bound for sparse or intermittent trusted paths. If all reliable paths to a target are rejected, the theorem applies only to the reachable part of the platoon, and that target cannot be reconstructed from communication alone.

B. Observer design procedure

The complete design can be implemented through the following steps.

- 1) Select a compact operating region $\Omega_x \times \Omega_u$ from vehicle limits and the receiving-side gate (??). Compute conservative bounds for the bicycle Lipschitz constants $L_{x,j}$ and $L_{u,j}$ on this region.

- 2) Choose the local observer gain $F_j(t)$ so that the zero-input local error recursion is uniformly locally exponentially stable. With full-state local measurements, verify the resulting LTV local-error recursion along the considered trajectory, for example using the parameter-dependent Lyapunov inequality in Lemma ??; frozen-time eigenvalue placement alone is not sufficient.
- 3) Fix the trust parameters using the nominal-data procedure in Section ??, then compute $\mathcal{VT}_{i,t}(t)$, $\mathcal{LT}_{i,j}(t)$, and the acceptance variables $\delta_{i,l}(t)$ and $\chi_{i0}^{(j)}(t)$ online.
- 4) Compute the anchor and neighbor weights from (??)–(??). Lemma ?? guarantees that the resulting correction is a nonnegative bounded mixture.
- 5) Check the contraction condition (??) or (??). If the check fails for a target, the implementation should increase the anchor budget when a reliable anchor exists, reduce neighbor reliance, hold the last trusted estimate, or mark the target temporarily unreachable rather than injecting unverified cooperative data.

C. Stability of the error

For target j , let $e_0^{(j)} = \hat{x}_0^{(j)} - x_j$, $e_i^{(j)} = \hat{x}_i^{(j)} - x_j$, and $e_{\mathcal{DO}}^{(j)} = \text{col} \left(e_1^{(j)}, \dots, e_N^{(j)} \right)$. The local and distributed nonlinear error recursions induced by (??) and (??) are

$$e_0^{(j)}(t+1) = f_b(\hat{x}_0^{(j)}, u_j) + F_j(y_j - h_j(\hat{x}_0^{(j)})) - x_j(t+1), \quad (52)$$

$$e_i^{(j)}(t+1) = f_b(\hat{x}_{i,c}^{(j)}, \hat{u}_j) - x_j(t+1), \quad i \in \mathcal{V}, \quad (53)$$

where $\hat{x}_{i,c}^{(j)}$ is the trust-weighted correction in (??); time arguments on its right-hand side are omitted for readability. The analysis is local on a compact operating region $\Omega_x \times \Omega_u$ with bounded speed and inputs, no yaw-chart discontinuity, and $|\delta| \leq \delta_{\max} < \pi/2$. On this region there are finite $L_{x,j}, L_{u,j} > 0$ such that

$$\|f_b(\xi, \mu) - f_b(\zeta, \nu)\|_* \leq L_{x,j} \|\xi - \zeta\|_* + L_{u,j} \|\mu - \nu\|_* \quad (54)$$

for all $(\xi, \mu), (\zeta, \nu) \in \Omega_x \times \Omega_u$. Conservative Lipschitz bounds and the local gain are computed from the trajectory Jacobians

$$\begin{aligned} A_i(t) &= \frac{\partial f_b}{\partial x} \Big|_{(\bar{x}_i, \bar{u}_i)}, & B_i(t) &= \frac{\partial f_b}{\partial u} \Big|_{(\bar{x}_i, \bar{u}_i)}, \\ C_i(t) &= \frac{\partial h_i}{\partial x} \Big|_{\bar{x}_i} = I_4. \end{aligned} \quad (55)$$

Throughout the distributed-error analysis, we use the block maximum norm

$$\|e_{\mathcal{DO}}^{(j)}(t)\|_{b,*} := \max_{i \in \mathcal{V}} \|e_i^{(j)}(t)\|_*. \quad (56)$$

The same block-norm convention is used for other stacked state vectors below. By Lemma ??, $M^{(j)}(t) = I_N - \tilde{\mathcal{L}}^{(j)}(t)$ is entrywise nonnegative and its i th row sum is $1 - w_{i0}^{(j)}(t)$.

Nonnegativity allows the triangle inequality to pass through the weighted correction: for every host i ,

$$\left\| \sum_{k=1}^N M_{ik}^{(j)}(t) e_k^{(j)}(t) \right\|_* \leq \sum_{k=1}^N M_{ik}^{(j)}(t) \|e_k^{(j)}(t)\|_* \leq (1 - w_{i0}^{(j)}(t)) \|e_{\mathcal{DO}}^{(j)}(t)\|_{b,*}. \quad (57)$$

Taking the maximum over i yields the induced mixing norm $\|M^{(j)}(t)\|_\infty = \max_i (1 - w_{i0}^{(j)}(t))$ used in the contraction arguments.

Stacking the trust-weighted correction and using the non-negative row-substochastic weights gives

$$\|e_{c,\mathcal{DO}}^{(j)}(t)\|_{b,*} \leq \|M^{(j)}(t)\|_\infty \|e_{\mathcal{DO}}^{(j)}(t)\|_{b,*} + \|w_0^{(j)}(t)\|_\infty \|e_0^{(j)}(t)\|_* + \|\rho^{(j)}(t)\|_{b,*}, \quad (58)$$

where $M^{(j)} = I_N - \bar{L}^{(j)}$, $w_0^{(j)}$ stacks the anchor weights, and $\rho^{(j)}$ is zero in the no-wrap Euclidean chart (or a bounded second-order retraction remainder).

The Jacobians in (??) provide the usual auxiliary linearized model for selecting F_j and checking conservative contraction constants, but the result below is stated for the nonlinear recursions (??)–(??).

If $a_0^{(j)}$ and $a_l^{(j)}$ corrupt the anchor and neighbor- l channels, respectively, the correction at host i contains

$$\chi_{\text{att},i}^{(j)}(t) = w_{i0}^{(j)}(t) a_0^{(j)}(t) + \sum_{l \in \mathcal{M}_i^{(j)}(t)} w_{il}^{(j)}(t) a_l^{(j)}(t). \quad (59)$$

By (??), this perturbation enters the nonlinear error recursion as an additive ISS input bounded by

$$\|d_{\text{att},i}^{(j)}(t)\|_* \leq L_{x,j} \|\chi_{\text{att},i}^{(j)}(t)\|_*. \quad (60)$$

Stacking these row-wise terms gives the additive input $d_{\text{att}}^{(j)}$. Its size is modulated by the trust weights. If $\|a_0^{(j)}(t)\|_* \leq \bar{a}_0$ and $\|a_l^{(j)}(t)\|_* \leq \bar{a}_N$ on $\mathcal{M}_i^{(j)}(t)$, then Lemma ?? gives

$$\|d_{\text{att},i}^{(j)}(t)\|_* \leq L_{x,j} \left(w_{i0}^{(j)}(t) \bar{a}_0 + \sum_{l \in \mathcal{M}_i^{(j)}(t)} w_{il}^{(j)}(t) \bar{a}_N \right) \leq \bar{A}_j \max\{\bar{a}_0, \bar{a}_N\}. \quad (61)$$

where $\bar{A}_j$ is finite on the compact region. Thus $\|d_{\text{att}}^{(j)}\| \leq \bar{d}_{\text{att}}$ for some finite $\bar{d}_{\text{att}}$; after rejection, the corrupted channel's term in (??) is zero.

For the nonlinear ISS proof, $d_0^{(j)}$ and $d_{\mathcal{DO}}^{(j)}$ denote the corresponding lumped inputs caused by model uncertainty, input mismatch, chart remainders, and bounded approximation terms. We assume these inputs and the additive attack contribution are bounded.

Assumption 3. *There are finite $\bar{d}_0$, $\bar{d}_{\mathcal{DO}}$, and $\bar{d}_{\text{att}}$ such that $\|d_0^{(j)}\| \leq \bar{d}_0$, $\|d_{\mathcal{DO}}^{(j)}\| \leq \bar{d}_{\mathcal{DO}}$, and $\|d_{\text{att}}^{(j)}\| \leq \bar{d}_{\text{att}}$. Hence $\|\tilde{d}_{\mathcal{DO}}^{(j)}\| \leq \bar{d}_{\mathcal{DO}} := \bar{d}_{\mathcal{DO}} + \bar{d}_{\text{att}}$.*

Lemma 4 (Local nonlinear observer ISS). *Assume that, on the compact operating region $\Omega_x \times \Omega_u$, the disturbance-free local nonlinear error recursion associated with (??) is uniformly*

locally exponentially stable. Equivalently, there exist $c_\ell > 0$ and $\lambda_\ell \in (0, 1)$ such that the zero-input local error satisfies the corresponding exponential bound for all admissible trajectories in the region. If Assumption ?? holds and the perturbed trajectory remains in the same region, then the local-observer error is ISS with respect to $d_0^{(j)}$. In particular,

$$\|e_0^{(j)}(t)\|_* \leq c_\ell \lambda_\ell^t \|e_0^{(j)}(0)\|_* + \frac{c_\ell}{1 - \lambda_\ell} \bar{d}_0. \quad (62)$$

A sufficient local check is the uniform exponential stability of the LTV local-error recursion

$$z(t+1) = (A_j(t) - F_j(t)C_j(t))z(t), \quad (63)$$

understood as an LTV system and not as frozen-time Schur stability of each $A_j(t) - F_j(t)C_j(t)$. A checkable sufficient condition is the existence of a bounded symmetric matrix sequence $P_j(t) = P_j(t)^\top$ and scalars $0 < \underline{p} \leq \bar{p}$ and $q > 0$ such that, with $A_{F,j}(t) := A_j(t) - F_j(t)C_j(t)$,

$$\underline{p}I \preceq P_j(t) \preceq \bar{p}I, \quad A_{F,j}(t)^\top P_j(t+1) A_{F,j}(t) - P_j(t) \preceq -qI \quad (64)$$

for all t , together with a sufficiently small invariant neighborhood on which the higher-order remainder of the nonlinear local-error map is uniformly quadratic.

Proof. Uniform local exponential stability of the zero-input recursion gives a transition bound $c_\ell \lambda_\ell^{t-k}$. Unrolling the perturbed local recursion and using Assumption ?? gives the geometric ISS estimate (??). The Lyapunov inequality (??) establishes uniform exponential stability of the LTV linearization; the uniformly quadratic higher-order remainder then gives uniform local exponential stability of the nonlinear zero-input recursion on a sufficiently small invariant neighborhood. ■

Lemma 5 (Nonlinear distributed-error contraction). *Suppose that the block contraction condition (??) below holds. If the local-observer error is ISS and Assumption ?? holds, then the nonlinear distributed error (??) is ISS with respect to $e_0^{(j)}$ and $d_{\mathcal{DO}}^{(j)}$. Under the additive attack model, the attacked distributed error is ISS with respect to $e_0^{(j)}$ and $\tilde{d}_{\mathcal{DO}}^{(j)}$.*

Proof. Use the componentwise error-size vector $E^{(j)}(t)$ defined before (??). Equations (??) and (??), together with the anchor and disturbance terms, give a componentwise recursion of the form

$$E^{(j)}(t+1) \leq L_{x,j} M^{(j)}(t) E^{(j)}(t) + b^{(j)}(t), \quad (65)$$

where $\|b^{(j)}(t)\|_\infty$ is bounded by finite gains multiplying $\|e_0^{(j)}(t)\|_*$ and $\|\tilde{d}_{\mathcal{DO}}^{(j)}(t)\|_{b,*}$. Define the block mixing product

$$\Phi_M^{(j)}(t, T_c) := M^{(j)}(t + T_c - 1) \cdots M^{(j)}(t). \quad (66)$$

Unrolling (??) over a block of length T_c and using (??) gives

$$\begin{aligned} \|E^{(j)}(t + T_c)\|_\infty &\leq G_{x,j}(T_c) \|\Phi_M^{(j)}(t, T_c)\|_\infty \|E^{(j)}(t)\|_\infty \\ &\quad + c_b \sup_{\tau \in [t, t+T_c-1]} \|e_0^{(j)}(\tau)\|_* \\ &\quad + c_b \sup_{\tau \in [t, t+T_c-1]} \|\tilde{d}_{\mathcal{DO}}^{(j)}(\tau)\|_{b,*} \end{aligned} \quad (67)$$

for some finite $c_b > 0$. Condition (??) contracts the homogeneous term by at most $\alpha_c < 1$. Iterating (??) gives a geometric ISS bound at the block times, and the bounded one-step gains extend it to the intermediate samples. Lemma ?? and Assumption ?? bound the inputs. In the nominal case, $d_{\text{att}}^{(j)} = 0$ and $\bar{d}_{\mathcal{DO}}^{(j)} = d_{\mathcal{DO}}^{(j)}$. ■

Combining Lemmas ?? and ?? gives the following result.

Theorem 1. Consider a vehicle platoon (??) over a directed weighted communication network $\bar{\mathcal{G}}^{(j)}$. Let (??) be the corresponding nonlinear distributed observer. Assume that the following conditions hold true:

- 1) *For all $j \in \mathcal{V}$, there exists a bounded gain $F_j(t)$ such that the local nonlinear observer error recursion (??) is uniformly locally exponentially stable on $\Omega_x \times \Omega_u$. A sufficient check is the uniform exponential stability of the LTV recursion (??) along the considered bounded bicycle trajectory, for instance through the parameter-dependent Lyapunov inequality (??) stated in Lemma ??.* Frozen-time eigenvalue placement of $A_j(t) - F_j(t)C_j(t)$ alone does not imply this property in the time-varying case.
- 2) *For all $i \in \mathcal{V}$, $w_{i0}^{(j)} \geq 0$ and $w_{il}^{(j)} \geq 0$ satisfy*

$$\sum_{l \in \mathcal{N}_i \cup \{0\}} w_{il}^{(j)}(t) \leq 1. \quad (68)$$

- 3) *There exist a window length $T_c \geq 1$ and a scalar $\alpha_c \in (0, 1)$ such that, uniformly over all admissible switches, the infinity norm induced by the block maximum norm (??) satisfies*

$$G_{x,j}(T_c) \left\| M^{(j)}(t + T_c - 1) \cdots M^{(j)}(t) \right\|_{\infty} \leq \alpha_c < 1. \quad (69)$$

Here $G_{x,j}(T_c)$ is the T_c -step bicycle prediction gain on the compact admissible region. A checkable sufficient condition for $T_c = 1$ is Lemma ??; for intermittent anchors or sparse graphs, Assumption ?? and Lemma ?? give the corresponding block check (??).

Then, the original nonlinear error dynamics (??)–(??) are locally ISS on $\Omega_x \times \Omega_u$. Under the additive attack model, the attacked estimation error is also locally ISS. In particular, there exist constants $c_0, c_1 > 0$ and $\lambda \in (0, 1)$ such that

$$\|e_0^{(j)}(t)\| \leq c_0 \lambda^t \|e_0^{(j)}(0)\| + c_0 \bar{d}_0, \quad (70)$$

$$\begin{aligned} \|e_{\mathcal{DO}}^{(j)}(t)\| &\leq c_1 \lambda^t \|e_{\mathcal{DO}}^{(j)}(0)\| \\ &\quad + c_1 \left(\|e_0^{(j)}(0)\| + \bar{d}_0 + \bar{\bar{d}}_{\mathcal{DO}} \right). \end{aligned} \quad (71)$$

Proof. Condition 1 and Assumption ?? give the local-observer ISS bound by Lemma ?. Lemma ?? gives Condition 2 and the nonnegative row-substochastic structure used in (??). Condition 3 gives the distributed block contraction in Lemma ?. Applying that lemma and taking $\lambda = \max\{\alpha_c^{1/T_c}, \lambda_\ell\}$ gives the stated bounds after the finite within-block, bounded-gain, and norm-equivalence constants are absorbed into c_0, c_1 . In the nominal case $d_{\text{att}}^{(j)} = 0$ and $\bar{\bar{d}}_{\mathcal{DO}} = \bar{d}_{\mathcal{DO}}$. ■

Remark 4. The result is local because the compact region excludes steering singularities, yaw-wrap discontinuities, and

unbounded velocities or inputs. The Jacobians in (??) provide conservative design constants, while the LTV local-error check must be performed along the trajectory rather than through frozen-time eigenvalues alone. Dense accepted-anchor graphs use $T_c = 1$; sparse switching graphs use the ordered windows of Assumption ?. Rejection drives the malicious input gain to zero.

VII. CONTAMINATION ROLLBACK UNDER DELAYED TRUST DECISIONS

Trust values are computed from consistency windows and therefore may lag behind the first corrupted packet. During this detection delay, falsified data can enter the distributed observer before the corresponding neighbor is removed from the trusted set. To limit this transient contamination, each host vehicle can optionally maintain a finite rollback buffer and re-evaluate the most recent observer steps once a neighbor is flagged.

The rollback module is useful when contamination is propagated through an otherwise honest intermediate vehicle. Consider an attacked transmitter l^* and two honest vehicles m and i . During the trust-detection delay, l^* broadcasts a falsified position or heading, and vehicle m uses this packet in its own distributed estimate before the trust score has fallen below the rejection threshold. Later, vehicle i receives the fleet estimate from m . From the direct perspective of vehicle i , the neighbor m may still appear reliable, but the estimate carried by m already contains upstream contamination from l^* . If the observer only sets future weights associated with l^* to zero after detection, the biased state that entered the recursion during the previous steps remains embedded in the current estimate and can continue to affect prediction. Rollback is introduced to address exactly this delayed-decision effect: when the unreliable source is identified, the host returns to a recent stored state, removes the newly rejected contribution from the buffered updates, and replays the observer with the local anchor and the remaining trusted neighbors. Thus, rollback converts a late trust decision into a retrospective correction rather than only a future communication cut.

Assume that neighbor l^* is declared unreliable at time t . For a rollback horizon K , host vehicle i stores, for each observer $\mathcal{DO}_i^{(j)}$ and each time $k \in \{t - K, \dots, t - 1\}$, the pre-update state $\hat{x}_i^{(j)}(k)$, the received neighbor estimates, the trust weights, the anchor estimate, and the input estimate $\hat{u}_j(k)$. The replay is executed as follows:

- 1) form the newly rejected set $\mathcal{M}$ from the trust decision at time t ;
- 2) reset a temporary replay state to the stored state at the beginning of the buffer;
- 3) replay the observer from $t - K$ to $t - 1$ using the archived packets, archived weights, and archived inputs, but excluding every source in $\mathcal{M}$;
- 4) replace the current distributed estimate by the replayed value.

During replay, the corrected state x_r is used to recompute the innovation terms

$$\tilde{\eta}_{il}^{(j)}(k; x_r) = w_{il}^{(j)}(k) (\hat{x}_l^{(j)}(k) - x_r), \quad l \in \mathcal{N}_i, \quad (72)$$

$$\tilde{\eta}_{i0}^{(j)}(k; x_r) = w_{i0}^{(j)}(k) (\hat{x}_0^{(j)}(k) - x_r). \quad (73)$$

When l^* is flagged, the host resets the temporary state to the buffer start,

$$x_r \leftarrow \hat{x}_i^{(j)}(t - K), \quad (74)$$

and replays the observer update for $k = t - K, \dots, t - 1$ while excluding the flagged neighbor, or a flagged set $\mathcal{M}$:

$$x_r \leftarrow f_b \left(x_r + \sum_{l \in \mathcal{N}_i \setminus \mathcal{M}} \tilde{\eta}_{il}^{(j)}(k; x_r) + \tilde{\eta}_{i0}^{(j)}(k; x_r), \hat{u}_j(k) \right). \quad (75)$$

Finally, the current distributed estimate is corrected by setting

$$\hat{x}_i^{(j)}(t) \leftarrow x_r. \quad (76)$$

The rollback memory cost is proportional to $K \deg(i)$ per observer and can be implemented with archived packets and weights. The anchor innovation $\tilde{\eta}_{i0}^{(j)}(k; x_r)$ is retained during replay so that removing a corrupted neighbor does not leave the correction step purely open loop.

Lemma 6 (Bounded rollback reset). Assume that the stored packets, archived weights, inputs, and replayed states remain in the compact admissible region used in Theorem ??, and that the rollback horizon K is finite. Then the reset

$$\zeta_i^{(j)}(t) = x_r - \hat{x}_i^{(j)}(t^-)$$

induced by (??) is bounded. If successive rollback triggers are separated by a dwell time $T_r > 0$, the reset sequence can be treated as an additional bounded impulsive input, and the local ISS boundedness conclusion of Theorem ?? is preserved with an enlarged ultimate bound.

Proof. For fixed finite K , the replay map is a finite composition of the clipped packet map, the bounded weighted correction step, and the bicycle prediction map on a compact set. This composition is bounded, hence $\zeta_i^{(j)}(t)$ is bounded. The dwell-time condition prevents infinitely many reset jumps from accumulating over a finite time interval. Between two reset times, Theorem ?? gives the usual ISS decay and bounded-input response; at a reset time, the bounded jump is added as another bounded input. Iterating over the dwell-time intervals gives the same ISS form with a larger constant that depends on the reset bound and T_r . ■

A. Pose propagation and relative-measurement re-anchoring

Rollback mitigates short-term contamination caused by delayed trust decisions, whereas anchoring provides a repeated pose reference that limits long-term propagation of accumulated offsets when a trusted anchor or relative measurement is available. This motivates the distinction between ordinary pose propagation and optional relative-measurement re-anchoring.

Let $p = [X, Y]^T$ and write the consensus-corrected state as $\hat{x}_{i,c}^{(j)} = \text{col} \left(\hat{p}_{i,c}^{(j)}, \hat{\psi}_{i,c}^{(j)}, \hat{v}_{i,c}^{(j)} \right)$. The usual observer prediction propagates the corrected pose as

$$\hat{p}_i^{(j)}(t+1) = \hat{p}_{i,c}^{(j)}(t) + T_s \hat{v}_{i,c}^{(j)}(t) \begin{bmatrix} \cos \hat{\psi}_{i,c}^{(j)}(t) \\ \sin \hat{\psi}_{i,c}^{(j)}(t) \end{bmatrix}. \quad (77)$$

Equation (??) is a motion update, not a position reset. If $\hat{p}_{i,c}^{(j)}$ is biased by contaminated cooperative data, then clean velocity

or heading only gives a cleaner motion vector from a biased starting point. The position offset remains.

Re-anchoring is an optional correction before prediction. If host vehicle i has an onboard relative measurement to target j , and this measurement passes the same validity and trust checks used in the local-trust layer, then the host can build an independent pose reference for j . With the measured relative position expressed in the global frame,

$$p_{\text{rel},i}^{(j)}(t) = \hat{p}_0^{(i)}(t) + r_{i,j}^{\text{meas}}(t), \quad (78)$$

where $\hat{p}_0^{(i)}$ is the host local-observer position and $r_{i,j}^{\text{meas}}$ is the measured relative position from i to j . This reference does not come from the received distributed estimate of target j ; it comes from host sensing and the host's own local pose.

The host may then blend the propagated base pose with this relative-measurement reference:

$$p_{b,i}^{(j)}(t) = (1 - \rho_{i,\text{rel}}^{(j)}(t)) \hat{p}_{i,c}^{(j)}(t) + \rho_{i,\text{rel}}^{(j)}(t) p_{\text{rel},i}^{(j)}(t), \quad (79)$$

$$\rho_{i,\text{rel}}^{(j)}(t) \in [0, 1],$$

and predict from the resulting base pose,

$$\hat{p}_i^{(j)}(t+1) = p_{b,i}^{(j)}(t) + T_s \hat{v}_{i,c}^{(j)}(t) \begin{bmatrix} \cos \hat{\psi}_{i,c}^{(j)}(t) \\ \sin \hat{\psi}_{i,c}^{(j)}(t) \end{bmatrix}. \quad (80)$$

The scalar $\rho_{i,\text{rel}}^{(j)}$ determines how strongly the relative measurement is used. The case $\rho_{i,\text{rel}}^{(j)} = 0$ gives the standard observer prediction, $0 < \rho_{i,\text{rel}}^{(j)} < 1$ gives a soft re-anchor, and $\rho_{i,\text{rel}}^{(j)} = 1$ resets the position base to the relative-measurement pose. If the relative measurement is missing, rejected, or outside the sensing range, then $\rho_{i,\text{rel}}^{(j)} = 0$ and the observer reduces to the trust-weighted prediction and rollback mechanism already analyzed above. Thus, re-anchoring is not used as an additional proof assumption; it is a practical add-on for cases where an independent onboard relative measurement is available and accepted.

VIII. SIMULATION AND PLATFORM RESULTS

This section reports three validation layers. The numerical campaign uses a five-vehicle platoon and 15 saved runs over local, global, and simultaneous local-global attacks to evaluate RMSE, trust response, source suppression, and the dominant attack cases. The QCar/LIMO validation then checks rollback under real packet timing and uses Case 2 trajectories to show closed-loop path effects. Finally, the control check uses trust as a gate between CACC and ACC and verifies positive projected spacing during the attack interval.

A. Numerical Simulation Setup

The numerical simulation uses five vehicles with all-to-all V2V communication; each vehicle runs the observer, and small modeling variations are treated as uncertainties. The main parameters are $N = 5$, $T_s = 0.01$ s, wheelbase $L_w = 2.8$ m, local-trust exponents $(\omega_p, \omega_v, \omega_\psi, \omega_u) = (1.4, 1.0, 1.2, 0.8)$, class-filter decay $\beta_{\text{trust}} = 0.5$, class confidence $C_{\text{tr}} = 0.2$, $q = 5$ trust classes, heading tolerance $\psi_{\text{th}} = 0.15$ rad, $\theta_{\text{min}} = 0.5$,

and observer-weight budgets $(\eta_0, \eta_N) = (0.4, 0.4)$, leaving at least 0.2 self-prediction mass.

For the direct-anchor case of Theorem ??, the contraction check uses the scaled infinity norm $\|z\|_* = \|D_x z\|_\infty$, $D_x = \text{diag}(0.01, 0.01, 1, 0.2)$, to account for mixed state units. On $v \leq 33$ m/s and $|\delta| \leq 0.35$ rad, the induced one-step bicycle gain is $L_{x,\max} = 1.066$. With accepted anchor mass $\underline{w}_0 = \eta_0 \theta_{\min} = 0.20$, Lemma ?? gives $\alpha = L_{x,\max}(1 - \underline{w}_0) = 0.853 < 1$ for the all-to-all accepted-anchor topology. If a target anchor is rejected, no reconstruction from communication alone is claimed; the implementation holds or marks the estimate as temporarily unreachable until an accepted anchored path returns.

B. Evaluation Metrics

The validation reports estimation accuracy, trust response, source-weight suppression, severe residual-error cases, roll-back behavior, and projected spacing preservation.

The main estimation metrics are RMSE, peak error, trust detection/recovery delay, and attacker-source influence. For state channel $q \in \{X, Y, \psi, v\}$ and vehicle i ,

$$\text{RMSE}_{q,i} = \sqrt{\frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} (q_i(t) - \hat{q}_i(t))^2}. \quad (81)$$

For yaw, the error is $\text{wrap}_\pi(\psi_i - \hat{\psi}_i)$. Saved-run attack-window RMSE is computed over $[10, 15]$ s for non-attacker entries. Detection delay is measured from 10 s until the mean trust assigned to the attacker falls below $\theta_{\text{op}} = 0.70$, and recovery delay is measured from 15 s until the same trace remains above θ_{op} for 0.5 s. Attacker-source influence is the neighbor or global-source weight assigned to the attacker, excluding direct self-weight; suppression delay is the first attack-window time at which this influence remains below 10% of its pre-attack mean for 0.5 s. The raw combined RMSE used for ranking is

$$\text{RMSE}_{\text{raw},i} = \text{RMSE}_{X,i} + \text{RMSE}_{Y,i} + \text{RMSE}_{\psi,i} + \text{RMSE}_{v,i}. \quad (82)$$

Because this sum mixes physical units, it is used only as a relative severity index within the same setup. False-positive trust rejection is the fraction of benign source-target samples below the selected trust threshold. The saved data do not contain the full post-attack state/input histories needed for estimation-error recovery, so recovery is reported only for trust and source-weight traces. The closed-loop check is limited to positive path-projected spacing during the attack interval.

C. Attack Scenarios

The numerical campaign is organized around five bicycle-state attack cases, summarized in Table ?. Each attack is active on the interval $[10, 15]$ s. The campaign contains 15 saved run files: local-channel corruption, global-channel corruption, and simultaneous local-global corruption, each crossed with attacker identities V1–V5. Within each file, the five attack IDs are evaluated as separate cases. Common random seeds are used for intermittent faults and packet drops, and the nominal true-state trajectory is fixed across runs. Unless stated

otherwise, the summaries below aggregate over state channels, attacker locations, attack cases, corruption channels, and non-attacker target vehicles.

TABLE II
FIVE BICYCLE-STATE ATTACK IDS USED IN THE NUMERICAL CAMPAIGN.

Case	Type	Target	Parameters
1	Bias	Planar (X, Y)	position bias = $[-5, 1]^T$ m
2	Faulty	Planar (X, Y)	position int. = 10 m, $p = 0.3$
3	Bias	Speed v	bias = -2.5 m/s
4	Faulty	Speed v	int. = 2.5 m/s, $p = 0.3$
5	Drop	All bicycle-state packets	$p_{\text{drop}} = 0.5$

D. Numerical Simulation Results

The trust layer detected every injected attack. On the $T_s = 0.01$ s grid, the mean detection delay is 0.0146 s, with case-level averages from 0.0075 s to 0.0500 s. Mean attacker trust decreases from 0.822 before attack to 0.207 during attack, a 0.615 drop or about 74.8%, and returns above $\theta_{\text{op}} = 0.70$ after 0.324 s on average. At the observer input, mean attacker-source influence decreases from 0.1386 to 0.0095, a 93.1% reduction, and is exactly zero in 88.5% of attack-window samples.

Table ?? compares the three corruption channels. Global-channel corruption is most damaging, with mean raw combined RMSE 0.1666, about 22.0% above local-only corruption and 18.7% above simultaneous local-global corruption.

The both-channel case remains closer to local-only than to global-only corruption because the added local inconsistency makes the compromised source easier to classify and quarantine, limiting its effective observer weight.

Table ?? ranks the five attack cases by mean raw combined RMSE over all channels and attackers. Case 2, the intermittent 10 m position fault, is most severe with mean raw combined RMSE 0.1737. The largest single vehicle-level impact also occurs in Case 2: global corruption from attacker V5 drives target V4 to raw combined RMSE 0.3175.

Fault-type attacks, especially the 10 m position fault, create the largest estimation degradation. Bias attacks produce smaller RMSE but stronger trust suppression because their persistent offsets are easier to recognize. Thus, low attacker trust confirms detection, whereas RMSE measures residual damage before and during weight suppression. The worst cell is used only as a campaign maximum, not as evidence that one attacker location is intrinsically more harmful; the robust conclusion is that global position-fault corruption dominates residual estimation error.

Table ?? reports attacker-as-source influence, excluding direct self-weight, so the values measure how much the corrupted vehicle can affect other distributed observers.

The both-channel case has the smallest residual source influence, with average attack-window source weight 0.0008 and zero-weight rate 99.1%, explaining why its RMSE remains close to local-only corruption. Global-only corruption is more damaging because it can remain partially trusted longer, with

TABLE III
ATTACK-CHANNEL SEVERITY OVER THE 15-RUN CAMPAIGN.

Channel	Mean raw RMSE	Max raw RMSE
Local	0.1366	0.1657
Both	0.1404	0.1833
Global	0.1666	0.3175

TABLE IV
ATTACK-CASE SEVERITY OVER ALL CHANNELS AND ATTACKERS.

Case	Meaning	Mean RMSE	Attack trust
2	P Faulty 10 m	0.1737	0.301
4	V Faulty 2.5 m/s	0.1481	0.313
5	DoS	0.1418	0.374
3	V Bias -2.5 m/s	0.1403	0.034
1	P Bias -5 m	0.1355	0.013

TABLE V
ATTACKER-SOURCE SUPPRESSION BY ATTACK CHANNEL.

Channel	Source wt.	Zero-rate	Supp. delay
Local	0.0118	83.9%	0.499 s
Both	0.0008	99.1%	0.044 s
Global	0.0159	82.6%	0.990 s

a larger attack-window source weight and slower suppression delay.

Overall, the trust layer identifies all saved attacks with mean delay 0.0146 s, reduces attacker-source influence by 93.1% on average, and assigns zero source weight in most attack-window samples. The remaining limitation is residual damage from global position-fault corruption before the source weight has fully collapsed, especially in Case 2; this condition should guide future tuning of trust thresholds, source suppression, and rollback activation.

E. QCar/LIMO Platform Rollback and Closed-Loop Trajectory Diagnostics

1) *Platform and Communication Stack*: The observer is also implemented on a mixed QCar/LIMO small-scale platform, where each vehicle runs the same estimation, communication, and control stack under real sensor timing, packet exchange, and controller execution. Vehicles broadcast local estimates, fleet estimates, trust summaries, and packet status. The receiving host checks physical and temporal consistency, updates trust weights, and runs the distributed observer with the local observer as an anchor. The numerical campaign uses labels V1–V5, whereas platform logs use V0, V1, and V2.

Peer vehicles exchange UDP packets, the trust observer converts consistency checks into adaptive fusion weights, and the fleet estimate is used for validation and control. The rollback comparison uses the existing cooperative controller; the spacing check uses the trust-blended cooperative/local implementation. The same attack channel supports matched no-rollback and rollback runs.

At vehicle level, both platforms share the normalized (X, Y, ψ, v) interface. The longitudinal branch projects position onto the road tangent and applies cooperative cruise

img/rollback_anchoring-eps-converted-to.pdf

Fig. 2. Case 1 local attack on QCar/LIMO: rollback and relative-pose anchoring under real-vehicle delay.

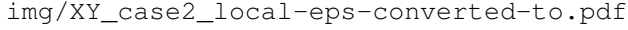
control, while the lateral branch uses cross-track and heading errors for steering.

2) *Rollback Evaluation Protocol*: The platform validation focuses on delayed-decision rollback rather than deterministic RMSE ranking because packet delays, asynchronous sensing, and execution jitter can let corrupted packets enter the observer before trust drops below the rejection threshold. The primary comparison is the same attack configuration without rollback and with finite-window rollback. After delayed detection, the host replays the recent observer window with the corrected trusted-neighbor set. This is mitigation, not perfect attack rejection, because the CACC-only controller may already have reacted to contaminated states.

Diagnostic clean-motion tests are separated from deployable observer operation. Clean-motion replay can use uncorrupted heading and speed in (??), but it keeps the estimator's current base position and exposes residual pose contamination. A hard re-anchor through (??)–(??) is realistic only with an accepted onboard relative measurement or another independently validated pose source; re-anchoring to a clean reference that bypasses the attack injector is used only as an ablation/debugging mode. The QCar/LIMO code, resources, and rollback video are provided in the online resources listed in the author footnote.

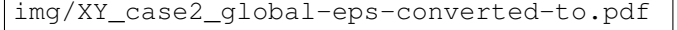
3) *Rollback and Anchoring Results*: To isolate rollback and anchoring, Fig. ?? compares three runs under the same attack configuration: trust-based estimation without rollback, with rollback, and with rollback plus relative pose anchoring. The time axis is shifted so $t = 0$ is attack onset. The target is V0, and the rejection signal is the direct attacked-source weight $w_0^{(0)}$ for the attacked direct/local channel.

Figure ?? shows that trust rejection alone is not sufficient once corrupted data have already entered the observer. Without



img/XY_case2_local-eps-converted-to.pdf

Fig. 3. XY trajectory under Case 2 local attack.



img/XY_case2_global-eps-converted-to.pdf

Fig. 4. XY trajectory under Case 2 global attack.

rollback, the direct attacked-source weight drops, but the contaminated state remains and the attack-window position RMSE reaches 3.77 m with peak error 3.87 m. Rollback replays the buffered observer updates after rejection; in this run, the rollback event occurs 0.058 s after attack onset and reduces the instantaneous post-prediction error from 1.97 m to 0.032 m, giving attack-window RMSE 0.168 m.

Rollback plus anchoring gives the smallest residual error because the relative pose anchor limits drift during the rejected-source interval. The attack-window RMSE decreases to 0.0218 m, with maximum error 0.0243 m. The three curves come from separate runs, so small timing differences in trust flags, rollback triggers, and direct-weight recovery should be interpreted as run-to-run timing variation rather than sample-synchronized traces.

4) *Closed-Loop Trajectory Diagnostics*: The same platform logs show how attacked estimates affect the driven path when supplied to the controller. Figures ?? and ?? plot Case 2 XY trajectories under local- and global-channel attacks, so they are closed-loop diagnostics rather than estimator-only plots.

In both Case 2 attacks, the corrupted source's trust and fusion weight decrease, shifting authority toward the self estimate and validated neighbors. Local attacks mainly reduce the direct/local component, whereas global attacks reduce the global-consistency component; the source-weight statistics in Table ?? quantify the resulting suppression.

F. Distributed State Estimation for Platoon Control

The trust scores are used both in the observer, where they limit unreliable V2V influence, and in the controller, where they gate cooperative CACC. Because a corrupted distributed estimate may affect the command before delayed trust correction, the longitudinal controller combines a local ACC channel

based on direct sensing of the immediate predecessor with a cooperative CACC channel based on distributed platoon estimates. A trust gate uses CACC when the fleet estimate is credible and shifts toward ACC when it is not.

The planar bicycle states are first projected onto the road tangent,

$$e_{\parallel}(k) = [\cos \psi_{\text{ref}}(k), \sin \psi_{\text{ref}}(k)]^{\top}, \quad (83)$$

$$s_i(k) = p_i(k)^{\top} e_{\parallel}(k), \quad (84)$$

where $p_i = [X_i, Y_i]^{\top}$. For follower $i \geq 2$, let $\mathcal{P}_i = \{1, \dots, i-1\}$ denote the cooperative predecessors. The equations use one-based platoon indexing; the experimental table keeps platform labels, where V0 corresponds to vehicle 1, V1 to vehicle 2, and so on.

The closed-loop validation uses the dominant Case 2 intermittent position-fault family, instantiated on the platform as a V0 attack schedule. This is a check of the dominant attack family, not a repeat of the exact worst attacker-target cell. Safety is evaluated by the minimum projected bumper-to-bumper spacing

$$s_{\min} = \min_{k \in \mathcal{T}_{\text{att}}} \min_{i=2, \dots, N} ((p_{i-1}(k) - p_i(k))^{\top} e_{\parallel}(k) - L). \quad (85)$$

The tested interval is longitudinally collision-free when $s_{\min} > 0$; the platform table uses 0.30 m as an experimental safety margin. This path-projected check does not prove full two-dimensional collision avoidance when lateral deviations are large.

The local adaptive-cruise branch uses only the measured predecessor gap and the time-headway target $d_{\text{ACC}} + h_{\text{ACC}} \max\{\hat{v}_0^{(i)}, 0\}$, so it remains available when vehicle-

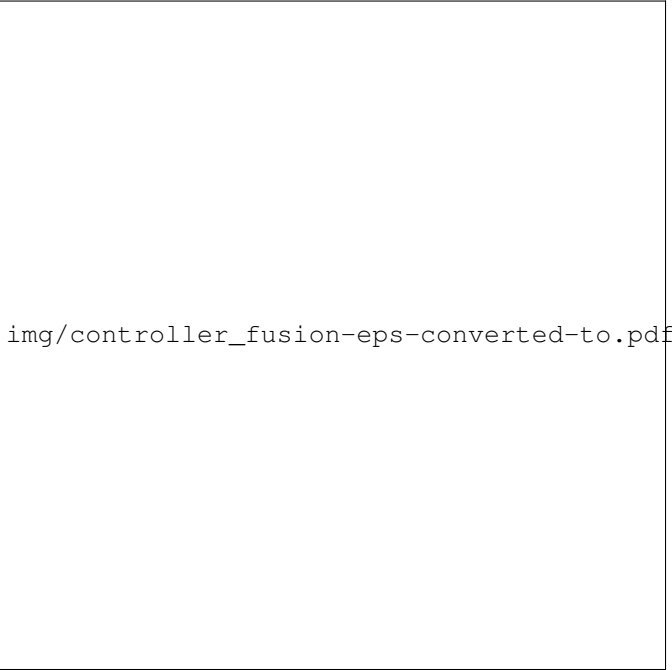


Fig. 5. Controller fusion spacing.

TABLE VI
ATTACK-WINDOW CLOSED-LOOP METRICS FOR THE PLATFORM V0
CASE 2 ATTACK

Host	Δt_{det} [s]	$\bar{\gamma}_{\text{att}}$	$\bar{w}_{\text{src}}(V0)$	s_{min} [m]	Gap RMSE [m]
V1	0.093	0.227	0.041	0.681	0.345
V2	0.090	0.894	0.048	0.637	0.171

to-vehicle data are rejected. The cooperative branch uses distributed predecessor position, speed, and heading errors:

$$u_{\text{coop},i}(k) = \sum_{j \in \mathcal{P}_i} [\kappa_s e_{s,ij}(k) + \kappa_v e_{v,ij}(k) + \kappa_\psi e_{\psi,ij}(k)], \quad (86)$$

where $e_{s,ij}$, $e_{v,ij}$, and $e_{\psi,ij}$ are the corresponding relative errors and $d_{i,j} = (i-j)(d + h\hat{v}_0^{(i)})$ for a homogeneous platoon. The trust-fused, filtered command is

$$\begin{aligned} u_{\text{target},i} &= u_{\text{loc},i} + \gamma_i(u_{\text{coop},i} - u_{\text{loc},i}), \\ \gamma_i &= \mathcal{V}\mathcal{T}_{i,i-1}, \\ u_i(k) &= u_i(k-1) + \tau_f(u_{\text{target},i}(k) - u_i(k-1)). \end{aligned} \quad (87)$$

Invalid or unavailable predecessor packets set $\gamma_i = 0$, selecting the local branch; credible data move the command toward cooperative control. Non-immediate corrupted sources are already attenuated by the observer weights. The experiment tests attack-window projected spacing, not string stability.

Table ?? summarizes two follower-host logs under the same V0 attack schedule. In both, the attacked V0 source weight falls below 0.05 on average within about 0.1 s. V1 shifts strongly toward local ACC because it directly follows V0, whereas V2 keeps more cooperation because its direct predecessor V1 remains consistent and V0 is already suppressed at the observer input. Both hosts preserve positive projected

spacing: the minimum gaps are 0.681 m and 0.637 m, above the 0.30 m margin. Thus, the reported gap RMSE reflects imperfect CACC tracking under attack, not loss of longitudinal collision clearance.

IX. DISCUSSION AND LIMITATIONS

The proposed method is designed to protect the platoon estimate and controller from unreliable incoming data, not to perform forensic reconstruction of a compromised vehicle. Once a neighbor is rejected by the trust layer, its information is no longer used as a cooperative correction source; accurate reconstruction of that vehicle then requires either a trusted independent measurement, a reliable local anchor, or a separate attack-identification module.

The ISS result also depends on bounded disturbances, bounded input mismatch, and sufficient trusted connectivity over time. If an attack removes all reliable paths to a target vehicle, no distributed observer can recover that target state from communication alone. Similarly, the rollback module improves delayed-decision recovery only when the required packets and weights remain inside the finite buffer and when trust-triggered resets do not occur arbitrarily fast. The current closed-loop evidence is limited to projected spacing preservation during the attack interval; string stability, sparse-topology operation, coordinated attacks, Monte Carlo noise/dropout behavior, and sensitivity to trust parameters require deeper validation.

X. CONCLUSION AND FUTURE WORK

This paper presented a trust-aware distributed observer for resilient state estimation in connected vehicle platoons. The proposed architecture combines local observer anchoring, distributed observer consensus, online trust evaluation, and an optional rollback module for delayed trust decisions. The trust scores define a time-varying trusted-neighbor set used directly in the observer-weight design, and sufficient ISS conditions were stated for bounded disturbances, input mismatch, and trust-induced switching. The evaluation is organized as a five-vehicle numerical RMSE campaign, a QCar/LIMO rollback validation for non-deterministic real-vehicle delays, and an estimation-based control check. Simulations under multiple attack cases show that the method can identify unreliable data and preserve bounded estimation errors. In the tested closed-loop scenario, the trust-modulated ACC/CACC controller preserves positive projected spacing during the attack interval.

Future work will focus on a deeper string-stability study of the closed loop, broader Monte Carlo validation, matrix-valued trust weights for channel-wise attack isolation, tighter dwell-time and contraction conditions for rollback-triggered resets, and stronger coupling between trust and control under dynamic communication topologies. Extending the QCar/LIMO validation to larger fleets and richer vehicle models is also a natural next step.